

Shubh Randeria

shubh@randeria.dev · +1 (512) 961-6743 · randeria.dev · github.com/acemavrick · linkedin.com/in/shubh-randeria

Education

Stanford University

B.S. in Aeronautics & Astronautics, Computer Science

Stanford, CA

June 2030

Awards

- National Merit Finalist (2026)
- GE Vernova STAR Scholarship (2026)
- HPE CodeWars 2025 & 2026, 2nd place, Advanced Division (Houston)

Experience

Product Development Intern (Stellar Agility Pte. Ltd.)

May 2025 – Aug 2025

github.com/robinlosey/songbook · apps.apple.com/us/app/on-wings-of-the-soul/id6748314818

Singapore

- Built and shipped *On Wings of the Soul*, an offline-capable iPad/iPhone songbook app (SwiftUI) pairing sheet music with recordings for 130+ songs, with search, curated collections, and favorites.
- Architected as a white-label template: catalog, content sources, branding, and theming live in centralized configuration, so new editions launch without touching app logic.
- Engineered over-the-air content updates from a versioned remote catalog with one-step rollback; atomic A/B Core Data swaps keep failed or interrupted updates from ever reaching users.
- Implemented a prioritized, bandwidth-aware download pipeline with Swift concurrency (sheet music first, audio deferred on metered networks, bounded parallelism, retries)

Projects

GPU Physics Simulations | Metal, Swift · github.com/acemavrick/simulations

Dec 2024 – Jul 2025

- Built a macOS suite of interactive, real-time GPU simulations of the 2D wave equation, electrostatics, and n-body gravity in Metal compute and fragment shaders; recently ported the n-body engine to Metal 4.
- Wrote finite-difference and RK4 integrators with per-pixel field visualization, GPU collision handling, and user-placed sources, walls, and charges.
- Designed adaptive sub-stepping that tunes compute passes per frame from measured GPU headroom, maximizing simulation speed at a steady 60 FPS.
- Rebuilt the wave simulation as a macOS screensaver with Metal Performance Shaders, explicit GPU teardown, and battery-aware rendering to withstand resource leaks in Apple's third-party screensaver host.
- Included audio visualizers: a real-time FFT spectrogram rendered as a 3D landscape (SceneKit) and a waveform view.

Mustang Bucks | Svelte, TypeScript, Firebase

May 2025 – Present

- Co-founded a digital currency platform replacing Cypress Ranch High School's analog reward system.
- Used daily by 3,400+ students and 200+ staff since September 2025.
- Led the frontend, UX, and admin dashboard (role-based access, roster imports, account management, exports) and built a comprehensive integration testing suite.
- Continue to maintain the platform in production, fixing bugs and shipping staff-requested features (e.g., bulk awards, per-transaction caps, etc.)

UIL-DL | Python, SvelteKit, Rust/Tauri · uildl.randeria.dev · github.com/acemavrick/uil-dl

Jul 2025 – Present

- Gives Texas students and coaches one-click access to ~600 UIL Academics contest packets otherwise scattered across yearly pages; shipped as a web app and desktop app, facilitating 1,500+ downloads since August 2025.
- Developed a structure-aware Python scraper that categorizes files by subject, level, and year, catching the nearly half of links naïve crawling missed, behind a remotely updatable index.
- Currently rebuilding the desktop app in Rust/Tauri with concurrency, rate-limiting, and type-safety improvements.